

Title & Setup

Title: Applied Data Science Capstone: SpaceX Falcon 9 First Stage Landing Prediction

Subtitle: IBM Data Science Professional Certificate Final Project

Author: [Your Name]

Project Repository: [GitHub Repository](#)

Executive Summary

Methodology Overview: Data was collected via the SpaceX API and web scraping Wikipedia. Data wrangling and exploratory data analysis (EDA) were conducted using Pandas, SQL, and data visualization tools (Matplotlib, Seaborn). Interactive analytics were built using Folium and Plotly Dash. Machine learning classification models were trained to predict landing success.

Key Results: The project successfully identified that launch site, orbit type, and payload mass strongly correlate with landing success. The predictive classification models achieved strong accuracy, with [Insert Best Model, e.g., Decision Tree] performing the best based on test data evaluation.

Introduction & Problem Statement

Project Background: SpaceX has revolutionized the space industry by reusing the first stage of its Falcon 9 rocket, bringing the cost of a launch down to \$62 million compared to competitors' upward of \$165 million.

Problem Statement: Determining if the first stage will land successfully allows us to determine the cost of a launch. The goal of this project is to predict the landing outcome of the Falcon 9 first stage to provide actionable pricing insights for a competing aerospace company.

Data Collection with APIs

Process Flowchart / Key Phrases:

- Execute **GET** request to the SpaceX API URL.
- Extract JSON payload containing flight data.
- Normalize JSON and convert to Pandas DataFrame.
- Filter for Falcon 9 launches and handle nested dictionaries to extract core, payload, and launchpad data.

Code Reference: [GitHub Repository](#)

Data Collection with Web Scraping

- **Process Flowchart / Key Phrases:**
 - Used the `requests` library to fetch the Wikipedia page for Falcon 9 launches.
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 - Parsed the HTML response using `BeautifulSoup`.
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 - Extracted data from HTML tables by iterating through `<tr>` and `<th>` tags.
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 - Appended scraped list data into a Pandas DataFrame.
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- **Code Reference:** [GitHub Repository](#)

Data Wrangling Methodology

Data Cleaning: Filtered missing values and calculated the mean of the **PayloadMass** column to replace nulls.

Outcome Calculation: Analyzed landing outcomes to categorize successful landings (True Ocean, True RTLS, True ASDS) as Class **1** and failed/unattempted landings as Class **0**.

Feature Engineering: Applied One-Hot Encoding to categorical variables such as **Orbit**, **LaunchSite**, **LandingPad**, and **Serial**.

Code Reference: [GitHub Repository](#)

EDA with Data Visualization

Purpose: To visually identify patterns, correlations, and anomalies between features and the target variable (landing success) before modeling.

Charts Utilized: Scatter plots were used to compare continuous variables (Flight Number, Payload Mass) against categorical variables (Launch Site, Outcome). Bar charts were deployed to evaluate the mean success rate across distinct Orbits.

Code Reference: [GitHub Repository](#)

EDA with SQL

Queries Performed:

- **SELECT** queries to isolate specific booster versions and launch sites.
- **GROUP BY** functions to aggregate total payload mass and average success rates per launch pad.
- **WHERE** and **BETWEEN** clauses to filter launches by date ranges.
- **ORDER BY** statements to rank successful landing counts by location.

Code Reference: [GitHub Repository](#)

Interactive Visual Analytics

- **Folium Map:** Generated an interactive geospatial map to visualize launch site locations. Plotted success (green) and failure (red) markers using MarkerClusters. Calculated proximity distances to coastlines, highways, and railways.
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- **Plotly Dash App:** Developed a reactive dashboard allowing users to filter launch sites via dropdowns and select payload ranges via a slider. This dynamically updated success pie charts and payload versus outcome scatter plots.
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- **Code Reference:** [GitHub Repository](#)

EDA Visualization Results

- **Scatter Plots:** The relationship between Flight Number and Launch Site showed that success rates improved significantly as flight numbers increased over time.
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- **Bar Charts:** Orbits such as ES-L1, GEO, HEO, and SSO displayed a 100% success rate, while the SO orbit had the lowest success rate.
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- **Yearly Trends:** A line chart plotting the year against the success rate confirmed a steady, progressive increase in landing success from 2013 to 2020.
- *(Remember to paste screenshots of your scatter plot, bar chart, and line chart here)*

EDA with SQL Results

- **Launch Sites & Payloads:** CCAFS SLC-40 handled the highest number of overall launches. Total payload mass varied significantly based on booster versions.
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- **Success Rates:** The queries successfully identified the exact date of the first successful ground pad landing and tracked successful drone ship landings.
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- **Rankings:** KSC LC-39A and CCAFS SLC-40 ranked highest for the most successful drone ship landings.

Add Slide 12: Folium Map Results

- **Launch Site Markers:** Visual clustering confirmed all launch sites are deliberately located near coastlines for safety over the ocean.
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- **Proximity Analysis Results:** Distance mapping revealed that launch pads are built extremely close to railways and highways for heavy transport logistics, but kept at a safe distance from major cities to minimize civilian risk.
- *(Remember to paste a screenshot of your Folium map showing the distance lines here)*

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Plotly Dash Results

- **Success Pie Charts:** The dashboard revealed that KSC LC-39A holds the highest overall success ratio among all individual sites (76.9% success rate).
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- **Payload vs. Outcome Scatter Plots:** Visualizations indicated that heavier payloads (above 4000 kg) launched from CCAFS SLC-40 tend to have higher failure rates compared to lighter payloads.
- *(Remember to paste screenshots of your Dash pie chart and scatter plot here)*

Predictive Analysis & Conclusion

- **Models Evaluated:** Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN).
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- **Evaluation & Confusion Matrix:** All models achieved an identical high accuracy on the test data (~83.3%). The confusion matrices confirmed accurate predictions of True Positives and True Negatives with very few False Positives.
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- **Best Model Explanation:** The Decision Tree classifier is considered the best model due to its high interpretability, making it easier to explain specific feature thresholds (like payload mass or orbit) to stakeholders.
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- **Conclusion & Innovative Insights:** First-stage reusability is highly predictable. The current SpaceX architecture guarantees a high probability of success for specific orbits (e.g., SSO). Competitors must account for an over 80% first-stage recovery rate when bidding for future commercial payloads.